Lesson 9 - More Models and Tuning

READ ME FIRST (theory). Then open the practice notebook.

Contents

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There are many models. The theory, then `practice.ipynb`. This lesson is an overview - do not worry about every detail.

Key words

Word	Simple meaning
Decision Tree	asks yes/no questions to decide
Random Forest	many trees together

Word	Simple meaning
KNN	copies the answer of the nearest examples
SVM	finds the best border between groups
regularization	keeps weights small to avoid overfitting
cross-validation	test on several splits for a fair score
Grid Search	tries many settings to find the best

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1. Decision Tree

A tree asks a series of yes/no questions about the features ("tenure < 6? contract = month-to-month?") until it reaches an answer. Easy to understand, but one tree can overfit.

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2. Random Forest

A **forest** is many decision trees together. Each tree votes; the forest takes the majority. It is more stable and overfits less than one tree.

Why it usually wins: the mistakes of single trees cancel out.

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3. KNN (K-Nearest Neighbors)

To predict a new point, KNN looks at the **K closest** known examples and copies their answer (the majority).

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4. SVM (Support Vector Machine)

SVM finds the best **border** that separates two groups, with the widest gap.

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5. Regularization (L1 / L2)

Regularization adds a small penalty for large weights, so the model stays simple and does not overfit. - **L2 (Ridge)** keeps weights small. - **L1 (Lasso)** can push some weights to exactly 0, removing useless features.

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6. Cross-validation

Instead of one train/test split, we try several different splits and average the scores. The result is more trustworthy.

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7. Grid Search - find the best settings

Models have **hyperparameters** (settings), like how deep a tree is. **Grid Search** tries many combinations and keeps the best one. No more guessing by hand.

Quick summary

- **Decision Tree** = yes/no questions. **Random Forest** = many trees, stronger.
- **KNN** = copy the nearest neighbors. **SVM** = best border.
- **Regularization** stops overfitting (L1 can drop features).
- **Cross-validation** = fairer score. **Grid Search** = find the best settings.

Now do this

Open `practice.ipynb`, train a Random Forest, compare it, and do the assignment.